Structure-predicted solvent responses enable simulation-free hydration free-energy prediction

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Abstract

Hydration free energy quantifies transfer from gas into water, but high accuracy can require explicit-solvent sampling and nuclear-quantum treatment for every new solute. Solvent-conditioned descriptors can enrich molecular learning, yet are often recomputed per query. We asked whether calculated, empirical and corrected solvation responses could instead become reusable, structure-predicted coordinates. SolvAI learns 15 such priors from benchmark-disjoint sources and combines them with molecular descriptors in an experimentally supervised endpoint. At inference, only a SMILES is supplied; no molecular dynamics, path-integral dynamics or probe calculation is run. In a preregistered matched five-fold comparison on the 85 neutral solutes used to establish ARROW, the priors reduce mean absolute error from 0.303 to 0.202 kcal mol^{-1}. This experimentally supervised endpoint reaches the accuracy scale of ARROW/PIMD8 (0.205 kcal mol^{-1}), used solely as a high-fidelity simulation reference. Five complete partitions give 0.207 $\pm$ 0.004 kcal mol^{-1}. Shuffling abolishes the gain, which survives chemical separation. On a prospectively frozen external molecule-disjoint cohort, the same contrast lowers MAE from 1.532 to 1.153 kcal mol^{-1} and remains in a strict 97-molecule source-disjoint subset, although larger absolute errors delimit applicability. Solvent-response information can thus become reusable molecular supervision rather than a per-query calculation.

Introduction

Hydration free energy is the reversible work of transferring a molecule from the gas phase into water. It connects molecular structure to partitioning, binding and reactivity, and is a stringent test of how a model balances electrostatics, polarization, dispersion, repulsion and solvent reorganization. Experimental values are available for many small neutral compounds, notably through FreeSolv, but predicting them with comparable precision remains difficult.[1;2]

Physical simulation approaches the problem by resolving the solvent response. In an alchemical calculation the solute–water interaction is varied along a coupling path, and the resulting ensemble averages are integrated or combined statistically.[3] More detailed Hamiltonians improve the intermolecular interaction; path-integral molecular dynamics (PIMD) can additionally represent nuclear delocalization.[4] These ingredients can be accurate, but they demand repeated sampling for each new molecule. The ARROW study made this accuracy–cost relation unusually sharp: on 85 neutral solutes, classical ARROW has an exactly reconstructed mean absolute error (MAE) of 0.785 kcal mol^{-1}, whereas eight-bead PIMD reduces it to 0.205 kcal mol^{-1}.[5] Subsequent learned-potential work reproduced ARROW water-pair interactions and quantum corrections within simulation, showing that parts of this description are learnable while retaining trajectory-based inference.[6;7]

Data-driven solvation models span several arrangements. Direct endpoint models map a graph, fingerprint or three-dimensional structure to a free energy, while Δ-learning and multi-fidelity models transfer information between calculated and experimental targets.[8–11] Graph models can also replace a particular implicit-solvent calculation by learning its output directly from molecular structure.[12] Physically inspired molecular descriptors can improve hydration prediction when they encode electrostatics, polarizability, accessible surface and conformational response.[13;14] Other models retain physical computation at prediction time: hybrid alchemical/ML methods correct a simulated free energy, ML-PCM learns from continuum-model components, 3D-RISM outputs have served as hydration thermodynamic descriptors, and implicitly perturbed Hamiltonians represent a molecule through its response to real or hypothetical solvents.[15–18] These studies show the value of solvent-conditioned representations, although an alchemical, continuum, integral-equation or ensemble calculation is still performed for a new molecule.

A complementary line of work learns the physical descriptor itself. The ml-QM-GNN architecture predicts quantum-chemical atom and bond descriptors before using them in a reactivity model.[19] Broader evaluations find that such descriptors help most when they are relevant to the endpoint and can be recovered accurately, and that a surrogate's hidden representation can sometimes be more useful than its explicit predicted values.[20;21] Most directly for hydration, physically inspired solute descriptors have been predicted from molecular graphs and passed to a second free-energy model, eliminating their per-query electronic structure calculation.[22] These developments establish the major architectural ingredients. They leave a distinct solvation question: can response information from calculated, empirical and corrected sources be consolidated into an explicit, structure-predictable layer that adds information beyond ordinary molecular features and experimental endpoint labels?

SolvAI tests this question using 15 response coordinates drawn from quantum-continuum, alchemical, empirical, residual-corrected and conformational sources. Independent structure-to-response models learn each coordinate across external data; a second model uses their predictions, together with molecular structure and experimental hydration labels, to learn the endpoint (Fig. 1). Descriptor prediction is therefore the means to test a broader scientific question: whether complementary solvent information remains useful when the responses are inferred from structure rather than recomputed or measured. PIMD-derived features were tested during development and were not retained. PIMD8 is therefore a demanding accuracy reference, not a teacher of the final model.

To isolate the response layer, we compare SolvAI with the same endpoint architecture, experimental labels, molecular representation, sample weights, folds and random seeds after removing only the 15 predicted priors. The layer reduces MAE from 0.303 to 0.202 kcal mol^{-1}; the gain disappears when molecule–response alignment is destroyed by shuffling. It persists when related chemistry is excluded from the complete endpoint training pool and when all ARROW experimental labels are removed. Together these controls test whether predicted physical response contributes transferable information beyond ordinary structure regression.

Results

Reusable solvent-response priors

The final response vector spans distinct approximations to solvation. One coordinate describes COSMOtherm water response from CombiSolv-QM; five Abraham axes encode polarizability, hydrogen-bonding and partition response; two coordinates are residual-corrected explicit- and implicit-solvent predictions; one represents SMD(water); and six summarize conformer reweighting and the distribution of water response in ConfSolv.[9;23–28] Every coordinate is generated by a surrogate from structure. The endpoint receives a 2,048-bit Morgan fingerprint, 217 RDKit descriptors and these 15 predictions.

The response sources differ deliberately in scale and physical resolution. MolSolv contributed 1,729,545 SMD(water) calculations, of which 350,359 structures remained in the confirmatory teacher after quality and identity filtering.[29] ConfSolv contributed 5,392,567 water conformer records; aggregation, target availability and standardized-identity filtering left 17,829 structures for the response model. These counts are not placed on a common axis because one source counts calculations and the other conformer records. Complete definitions appear in Supplementary Table 1 and source-specific provenance in Supplementary Fig. 2.

An expanded identity audit changed the original release analysis. Exact connectivity had already been removed from every supervised external source, but conservative fragment, uncharging and canonical-tautomer standardization found 2 equivalent CombiSolv-QM records, 32 MolSolv records and 22 ConfSolv records. We removed them, refitted the three affected teachers while preserving all remaining source split memberships, and repeated every endpoint control reported below. The 1,280-label experimental endpoint pool contained no exact or standardized benchmark identity. This corrected analysis is the sole source of the main numerical results.

A matched test of physical information

Removing only the predicted responses from the frozen endpoint gives an OOF MAE of 0.303 kcal mol^{-1} (Fig. 2a). Adding the compact response block lowers the error to 0.212; adding SMD(water) gives 0.207; and the complete stack reaches 0.202 kcal mol^{-1}. The last value lies on the same accuracy scale as ARROW/PIMD8 (0.205 kcal mol^{-1}), although the paired comparison does not establish superiority over PIMD8.

The response advantage is resolved at molecule level. Full SolvAI improves the absolute error for 53 of 85 solutes and reduces MAE by -0.101 kcal mol^{-1} relative to the matched endpoint (95% paired-bootstrap interval, -0.215 to -0.020; Fig. 2b). The empirical and residual-corrected block is independently positive. Computation-derived, SMD and ConfSolv blocks are weaker in isolation, but their complete combination is positive (Fig. 2c). This is evidence of complementary response information, not an additive decomposition of hydration free energy. Molecule-level prediction, residual and paired-change diagnostics are shown in Supplementary Fig. 1.

Arbitrary additional columns do not reproduce the result. Across five independent permutations of the 15-column block, the fixed-partition mean is 0.306 kcal mol^{-1}, close to the 0.303 matched control and far from aligned SolvAI. The paired difference between aligned and mean-shuffled priors is -0.104 kcal mol^{-1} (95% interval, -0.214 to -0.025). Alignment between a molecule and its predicted response is therefore necessary for the observed gain.

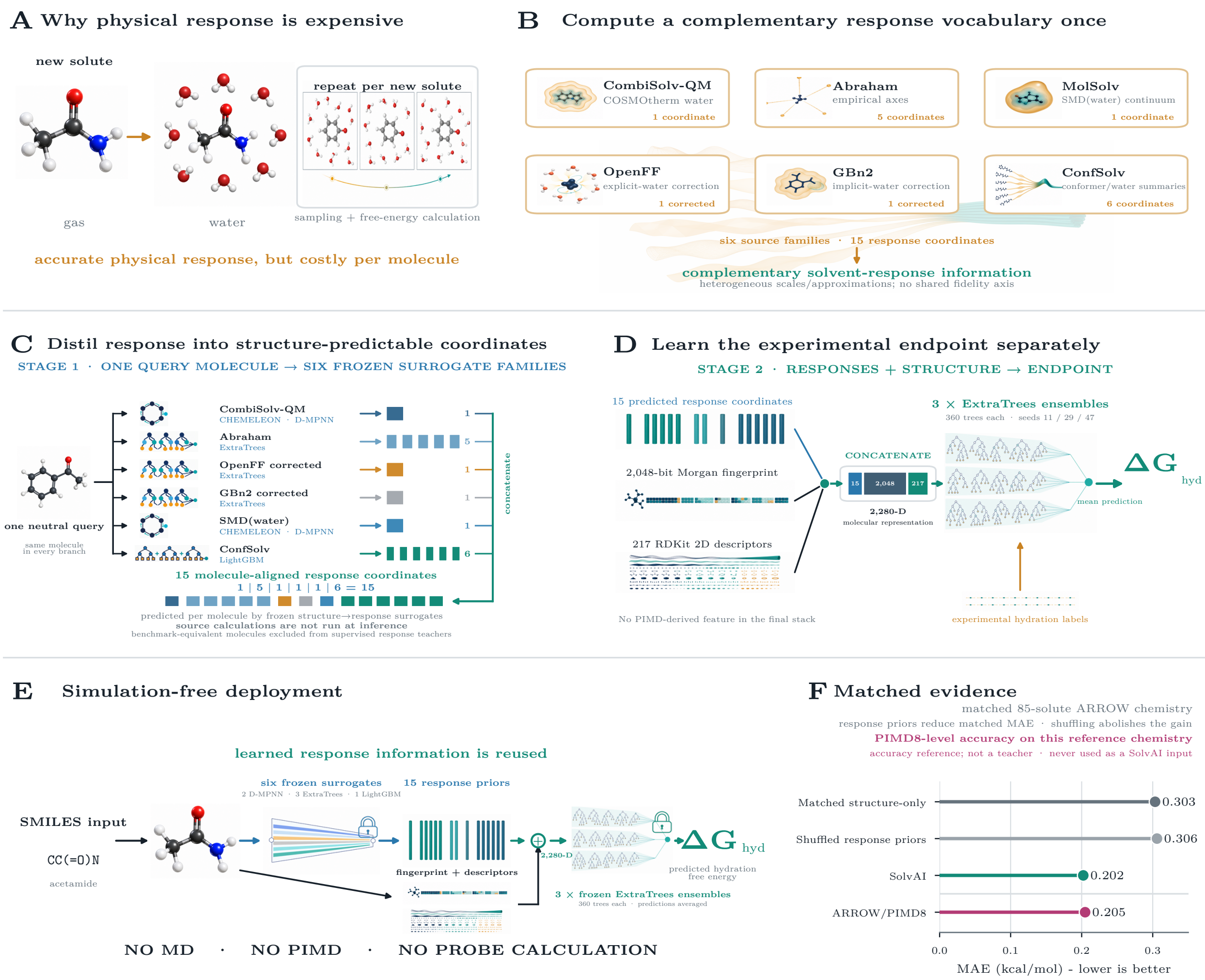

Figure 1: Physical response becomes reusable molecular supervision. External calculations and measurements describe complementary aspects of transfer into water, including continuum response, conformational reweighting and alchemical interaction response. In stage 1, structure-to-response surrogates learn 15 compact coordinates. In stage 2, their predictions and molecular descriptors are fitted to experimental hydration free energies: the 15 priors, 2,048 Morgan bits and 217 RDKit descriptors are concatenated once into a 2,280-dimensional representation, which is evaluated by three ExtraTrees ensembles and averaged. Deployment repeats only the learned mappings: a SMILES string produces a hydration free energy without MD, PIMD or a probe calculation. ARROW/PIMD8 is shown separately as an accuracy reference and is not a retained teacher.

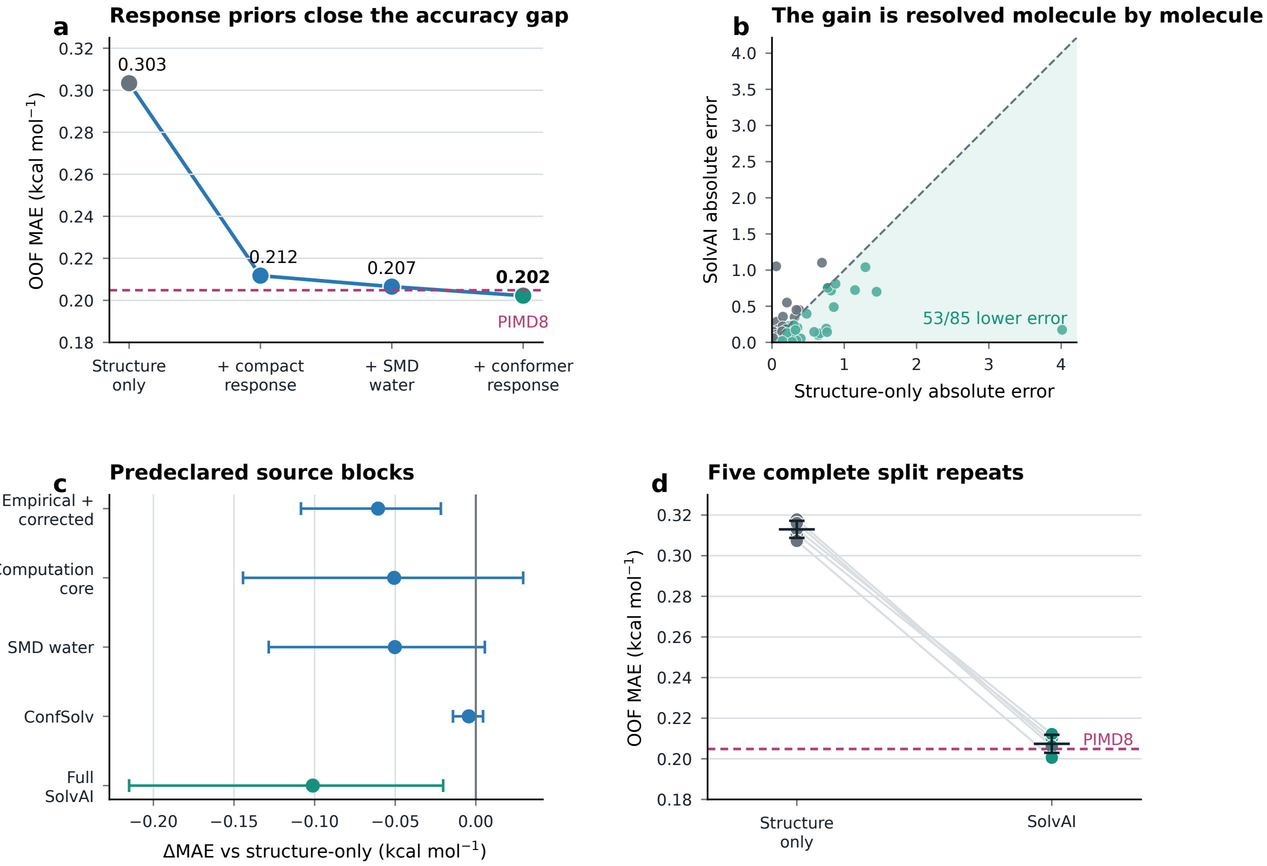

Figure 2: Molecule-aligned response priors improve a matched hydration endpoint. **a**, Predeclared fixed-partition progression. Each endpoint uses the same experimental labels, descriptors, architecture, weights, folds and seeds; only the response block changes. The dashed magenta line marks ARROW/PIMD8 on the same 85 solutes. **b**, Paired absolute errors for the matched structure-only endpoint and full SolvAI. Shading marks lower SolvAI error. **c**, MAE change for predeclared response blocks relative to structure only; bars are 95% molecule-bootstrap intervals. **d**, Paired results for five complete split repeats. Thin lines join methods within the same partition; black marks show mean ± sample s.d.

Partition choice changes the absolute point estimate but not the controlled advantage. Across five complete five-fold partitions, matched structure-only prediction gives 0.313 kcal mol^{-1} on average, whereas SolvAI gives 0.207 ± 0.004 kcal mol^{-1}; SolvAI is better in all five partitions (Fig. 2d). The appropriate headline is PIMD8-level accuracy on this reference chemistry, not uniformly sub-0.20 performance. Changing the prospectively frozen weight of available ARROW outer-training labels from 3 to 1 leaves the contrast essentially unchanged: 0.310 for structure only and 0.206 kcal mol^{-1} for SolvAI (paired change -0.103 kcal mol^{-1}; 95% interval, -0.219 to -0.020; Supplementary Table 9).

Transfer beyond nearby labels

Random molecular folds can leave close chemical neighbours in the external endpoint pool. We therefore applied each separation rule to all endpoint-supervised molecules, not only the 85 reference solutes. For each outer test partition, public and ARROW training rows in the held-out family, scaffold or molecular cluster were removed. A complementary analysis excluded every training row at or above a prescribed Morgan similarity to any test molecule.

The response-prior advantage survives all predeclared regimes (Fig. 3). At a maximum allowed train–test similarity of 0.70, MAE falls from 0.343 to 0.211 kcal mol^{-1}; cluster separation lowers it from 0.349 to 0.216. Scaffold and functional-family separation are much harder, but the same comparison favours SolvAI: 0.728 to 0.376 and 1.262 to 0.468, respectively. Every paired 95% bootstrap interval lies below zero. Response priors therefore retain value away from nearby endpoint labels, even where absolute endpoint accuracy is not yet sufficient. Molecule-level family errors and the four global separation regimes are expanded in Supplementary Fig. 5.

We next removed all 85 ARROW experimental labels from endpoint training. Using only the same 1,280 external labels, the matched structure model gives 0.385 kcal mol^{-1} and SolvAI gives 0.257 kcal mol^{-1} (paired change -0.128; 95% interval, -0.246 to -0.037). This transport test is not an independent external

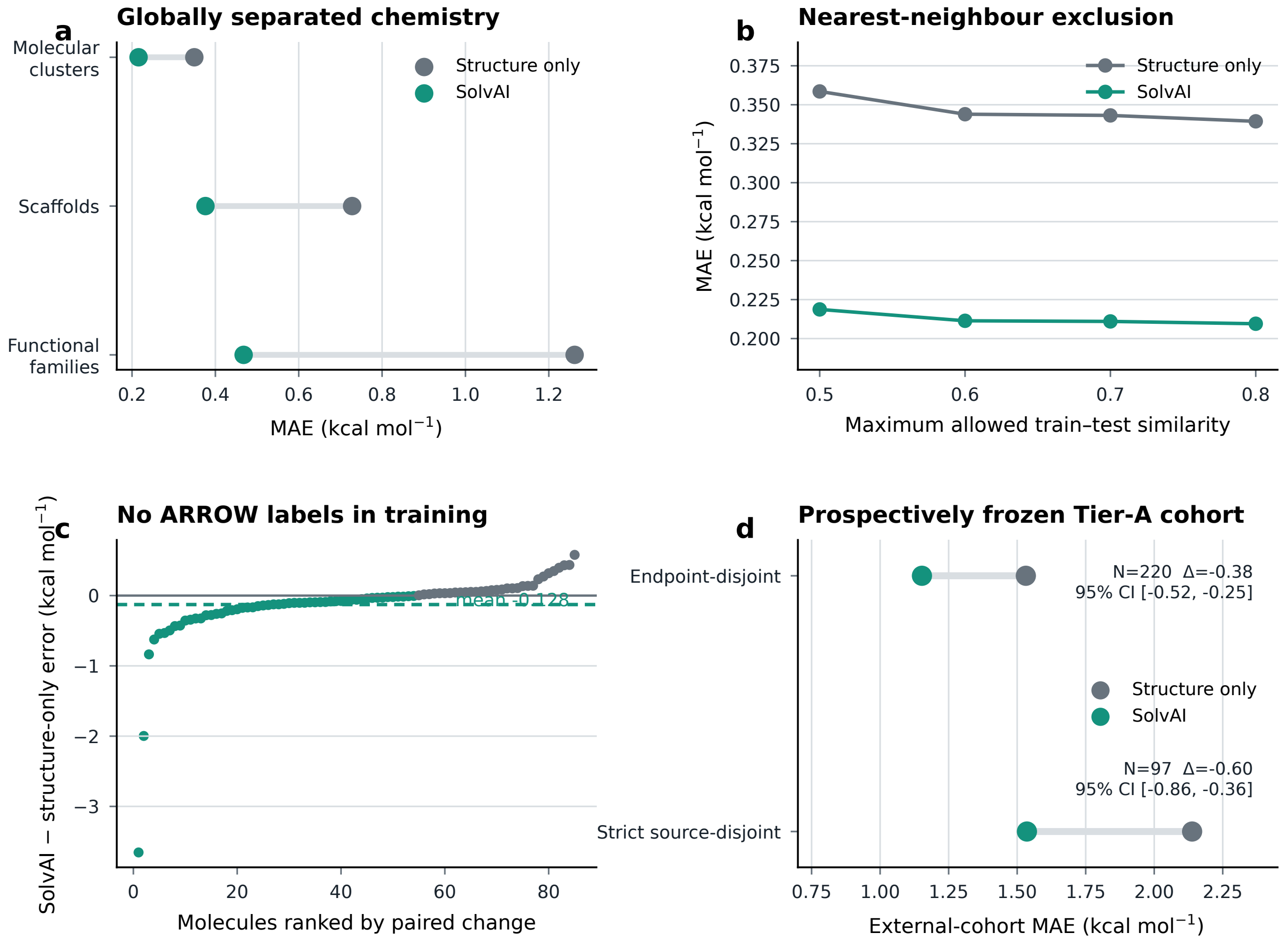

Figure 3: The response-prior advantage survives chemical separation. **a**, Matched MAEs after removing every endpoint training label in the held-out molecular cluster, scaffold or functional family. **b**, Sensitivity exclusion removes every training row at or above the indicated maximum Morgan similarity to a test molecule. **c**, Molecule-level change in absolute error when no ARROW experimental label is used for training; negative values favour SolvAI. **d**, Matched performance on the prospectively frozen Tier-A external cohort and the subset also absent from every response-teacher source; intervals summarize paired absolute-error differences, not the plotted MAEs. All response teachers use the standardized-equivalent exclusions described in Methods.

benchmark—the method was developed in the context of ARROW chemistry—but it shows that response priors retain value without fold-local adaptation to the reference set.

Finally, we evaluated a prospectively frozen external cohort derived from original measurements in Sander's Henry-law compilation. Eligibility was determined without model errors. Of 221 pre-existing candidates, one was excluded because an independent registry audit had already established a name–CAS–structure conflict. The remaining 220 neutral solutes have no exact or standardized-equivalent identity in any endpoint label used to fit the deployment models. On this endpoint-disjoint cohort, MAE falls from 1.532 to 1.153 kcal mol^{-1} (paired change -0.379; 95% interval, -0.517 to -0.248; Fig. 3d). Because some molecules occur in response-teacher source tables, we also froze a stricter subset of 97 molecules absent from all six response sources. The advantage remains: 2.138 to 1.536 kcal mol^{-1} (paired change -0.603; 95% interval, -0.863 to -0.359). The larger absolute errors show that this result extends the response-layer contrast, not the PIMD8-level accuracy claim, beyond ARROW-85.

What makes physics transferable

The benefit is not a generic consequence of adding physically motivated outputs. Compact ConfSolv response summaries transfer better than graph or feed-forward latent representations trained on the same source (Fig. 4a). Other candidate signals—including additional force-field diagnostics, pair-interaction data and learned implicit-solvent hierarchies—were non-improving in matched campaign screens (Supplementary Fig. 3). These results do not rank the underlying physical methods. They test whether each representation can be inferred from structure accurately enough, and with sufficiently non-redundant information, to improve the endpoint.

Short PIMD2 calculations at coupling values $\lambda = 0.1$, 0.5 and 0.9 make this constraint explicit. Structure-only heads captured broad response trends, but component MAEs ranged from 1.27 to 5.20 kcal mol^{-1} (Fig. 4b).

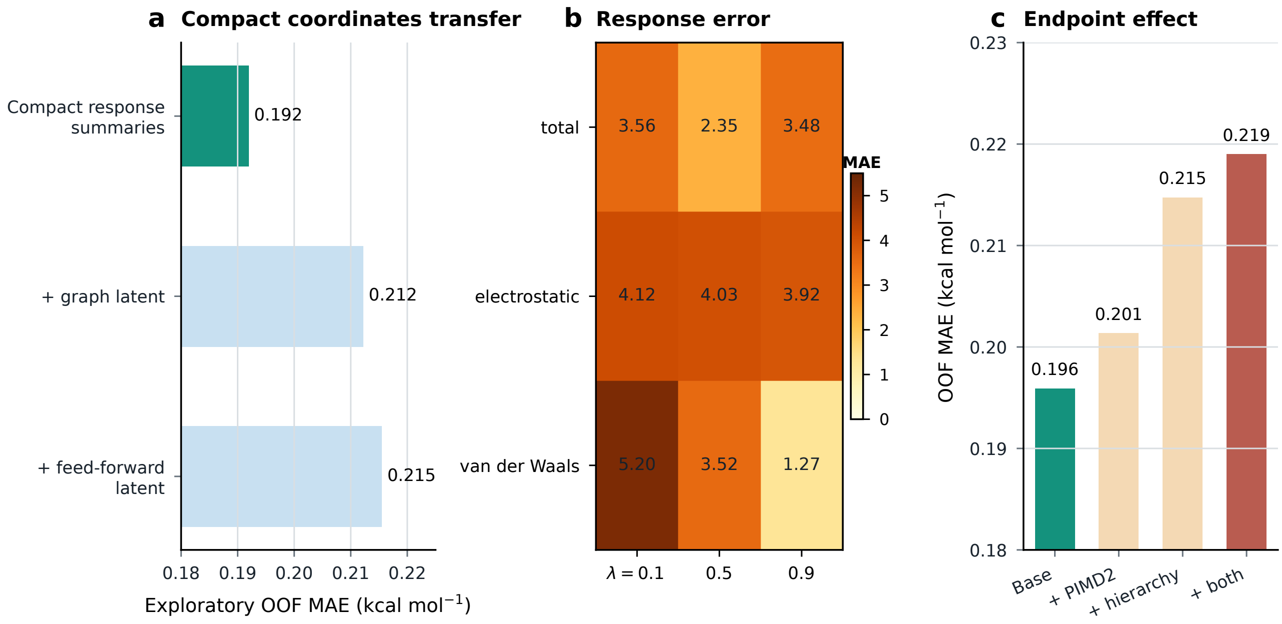

Figure 4: Transfer is limited by structure-to-response accuracy. a, Exploratory matched screen comparing compact ConfSolv summaries with graph and feed-forward latent coordinates from the same source. **b**, OOF error of structure-predicted PIMD2 total, electrostatic and van der Waals $\mathrm{d}H/\mathrm{d}\lambda$ responses. Values are kcal mol^{-1}. **c**, Consequence of adding those predicted high-fidelity responses to the matched campaign endpoint. These analyses explain a bottleneck; none of the PIMD-derived blocks is retained in SolvAI.

Adding the predicted PIMD2 response worsened the matched campaign endpoint from 0.196 to 0.201 kcal mol^{-1}; a predicted classical–NQE–PIMD hierarchy reached 0.215, and their combination 0.219 (Fig. 4c). Direct integration of the predicted response curve yielded 1.51 kcal mol^{-1} (Supplementary Fig. 4). No PIMD-derived coordinate is present in the final model.

The experiments suggest a three-part transfer condition. A response must be relevant to the endpoint, learnable from the deployed molecular representation, and non-redundant with the other priors. Broad physical fidelity alone does not guarantee useful distillation. Here, compact solvent-response coordinates learned from much larger external sources satisfy that condition more effectively than sparse, high-fidelity alchemical targets.

Discussion

SolvAI makes solvation-response information reusable rather than emulating one trajectory engine or one calculated endpoint. Quantum-continuum, alchemical, empirical, residual-corrected and conformational sources define a low-dimensional response vocabulary; structure surrogates reproduce that vocabulary; and an experimentally supervised endpoint learns how the predictions combine for hydration. The matched control and shuffled-prior experiment show that the benefit is not attributable to a larger feature matrix alone. Global separation and zero-ARROW transfer further show that it is not confined to close endpoint-labelled neighbours.

Predicted physical descriptors, Δ-learning and computed-to-experimental transfer are important precedents for this design.[8;9;19;22] SolvAI brings these ideas together through independently trained surrogates spanning distinct descriptions of solvation, then isolates their collective contribution under matched and destructive controls. The two-stage pattern is established. The unresolved question is whether complementary solvent responses supply distinct information when each is inferred from structure rather than recomputed or measured.

The transfer criteria are also consistent with systematic studies of quantum descriptors, which emphasize endpoint relevance and accurate inference.[20] The present controls show that a third condition also matters—non-redundancy with the deployed molecular representation. Recent work found that hidden surrogate representations can outperform explicit predicted descriptors in some small-data tasks.[21] Here, by contrast, compact ConfSolv response summaries outperformed graph and feed-forward latent coordinates from the same source. The ordering is task-dependent: the useful representation depends on which physical information survives structure prediction and remains complementary to the endpoint features. The transferable signal is therefore not "simulation" in the abstract, but an aligned, learnable representation of solvent response.

The evidence also sets a clear boundary. The primary reference contains only 85 neutral solutes, and the method was developed in its context. The confirmatory five-partition mean is 0.207 kcal mol^{-1}, so the evidence supports an accuracy scale rather than a threshold claim. Absolute error rises to 0.376 and 0.468 kcal mol^{-1}

when all endpoint labels sharing the test scaffolds or functional families are removed. The frozen model is not validated for ions, salts, biomacromolecules or substantially larger chemical space. Nor is this a generic FreeSolv ranking: 80 of the 85 reference identities occur in FreeSolv, and FreeSolv-related records contribute to endpoint training only after benchmark identity exclusion.

The external cohort sharpens this boundary rather than erasing it. Predicted response coordinates retain a statistically resolved advantage on both 220 endpoint-disjoint molecules and the 97-molecule response-source-disjoint subset, but full-model MAE is 1.153 and 1.536 kcal mol^{-1}, respectively. The present release also has no calibrated per-query applicability-domain or reliability score. It should therefore be treated as a validated predictor for the neutral small-molecule domains represented here, not as an assurance of uniform accuracy on arbitrary neutral chemistry.

The negative high-fidelity result is informative. To distil alchemical response directly, the model needs enough protocol-matched examples to predict component curves at substantially better than the present 1.27–5.20 kcal mol^{-1}. A focused set of roughly 250–500 molecules spanning amides, aromatics, ethers, acids and alkanes, with complete λ-resolved total, electrostatic, polarization and dispersion/repulsion response and paired classical/PIMD values, would test that next step. It is a more specific requirement than simply collecting more endpoint labels.

The broader opportunity is to treat costly computation as scientific supervision with a lifetime beyond one query. Once a useful response is represented across enough chemistry, its information can be predicted from structure and reused at negligible marginal simulation cost. Hydration provides a demanding demonstration because the accuracy reference includes explicit solvent and nuclear quantum effects. The same principle may apply wherever physical computation exposes transferable response coordinates that a deployed model can learn.

The packaged implementation is not a millisecond predictor: on the release host its warm CPU median is 15.29 s for one molecule and 15.82 s for a batch of 32 (0.494 s per molecule). Its computational advantage is the removal of per-query physical simulation. Comparisons with the published 15-window PIMD8 workload concern total work on different hardware, not a same-hardware kernel benchmark.

Methods

Reference set and targets

The primary evaluation uses the 85 neutral organic solutes reported in the ARROW hydration analysis at 298 K.[5] One water condition was retained per connectivity. The released benchmark table contains canonical structure, experimental hydration free energy, experimental uncertainty where available, classical ARROW, PIMD4 and PIMD8 values, and all fold assignments. The exact reconstructed PIMD8 MAE is 0.205 kcal mol^{-1}; the source article reports approximately 0.20 kcal mol^{-1}.

Endpoint labels and features

The external endpoint pool contains 1,280 neutral, benchmark-disjoint experimental hydration labels: 1,147 connectivity-resolved records from a FreeSolv/CombiSolv-Exp merge and 133 SoluteML dGsolvDB3 records supported by at least two source measurements.[2;24;30] Each endpoint input has a 2,048-bit radius-2 Morgan fingerprint and 217 RDKit two-dimensional descriptors. Non-finite values are handled by a median imputer fitted inside each endpoint fit.

Response surrogates

Fifteen response coordinates are generated by six artifact families (Supplementary Table 1). CHEMELEON-initialized D-MPNNs predict CombiSolv-QM/COSMOtherm and MolSolv SMD(water) targets.[31] ExtraTrees models predict five Abraham axes and the physical and residual components of corrected OpenFF and GBn2 responses. LightGBM models predict six ConfSolv conformer and response-distribution summaries. The latter are derived from the H_2O subset of ConfSolv. The final coordinate list, target units, source sizes, transformations and inference calculations are given in Supplementary Table 1. PIMD-derived targets are absent from this list.

The endpoint is an ensemble of three `ExtraTreesRegressor` pipelines,[32] each with 360 trees, maximum feature fraction 0.7, minimum leaf size 2, no bootstrap and random seeds 11, 29 and 47. Predictions are averaged. Public endpoint rows have sample weight 1 and available ARROW outer-training rows weight 3. The matched structure-only model is identical except that the 15 response columns are removed.

Identity standardization

RDKit 2026.03.5 was used to generate canonical isomeric SMILES, full InChIKeys and connectivity blocks.[33] The confirmatory audit additionally applied MolStandardize cleanup, largest-fragment selection, uncharging and canonical tautomer enumeration. Exact and standardized benchmark equivalents were removed from every supervised external source. The audit found no such match in the endpoint pool. It identified 2 CombiSolv-QM, 32 MolSolv and 22 ConfSolv source records beyond the original exact-connectivity exclusion. The affected teachers were refitted with all remaining molecules held in their original train, validation or test split. Full pair lists, identifiers and split-verification hashes are released.

Confirmatory evaluation

The analysis protocol was committed before valid confirmatory model fitting. The primary partition is the existing five-fold assignment on the 85 molecules. Five complete shuffled K-fold partitions use seeds 314159, 271828, 161803, 141421 and 173205. In each outer fold, all 1,280 external labels and the available ARROW outer-training labels are fitted; the test target is absent. Feature blocks, ExtraTrees hyperparameters, sample weights and model seeds are identical across all comparisons.

A separately committed one-factor sensitivity repeated the fixed folds after changing only the available ARROW outer-training weight from 3 to 1. It retained all source rows, teachers, features, hyperparameters and seeds. Its paired interval used the same 100,000-resample convention.

For the negative control, the 15 response columns are permuted as one block, thereby preserving their joint distribution while destroying molecule alignment. Public, outer-training and outer-test response rows are permuted separately within each fold. Five permutation seeds (88001–88005) are evaluated for the fixed partition and each of the five repeats.

Chemical separation

Separation is applied to the complete endpoint-supervised pool. Functional families are assigned by a deterministic SMARTS precedence; five outer folds are formed by group. Bemis–Murcko scaffolds are canonicalized, with acyclic structures grouped by primary family. Taylor–Butina clusters use radius-2, 2,048-bit Morgan fingerprints at Tanimoto similarity 0.70. For each outer test fold, every public or ARROW training row sharing a held-out group is removed. A separate nearest-neighbour analysis removes every training row with similarity at or above 0.50, 0.60, 0.70 or 0.80 to any test molecule. No threshold was selected by performance. Exact SMARTS, assignments, training counts and maximum retained similarities are released as Supplementary Data 3.

Zero-ARROW transfer

The frozen endpoint configuration was fitted once to the 1,280 external experimental labels without any ARROW experimental value. Both matched structure-only and full response-prior models then predicted all 85 ARROW molecules. This analysis measures transport without reference-set endpoint adaptation; it is not described as an independent external test because model development used the reference chemistry.

Tier-A external validation

The candidate cohort was frozen independently in an earlier ASP-19/GB study from Sander Henry-law compilation v5.0.0 original-measurement records.[34] Intrinsic water constants at 298.15 K were converted using $K_c = H_s^{cp}RT$ and $\Delta G_{\text{hyd}} = -RT \ln K_c$, corresponding to 1 M ideal gas and 1 M ideal-dilute aqueous standard states. Qualification preceded SolvAI evaluation and required one connected neutral solute, supported elements, compatible target metadata and no exact, fragment-parent, uncharged-parent or canonical-tautomer equivalent in the 1,280 external or 85 ARROW endpoint labels. One previously documented registry conflict was excluded, leaving 220 molecules. A strict 97-molecule subset additionally excludes any identity-equivalent occurrence in all six supervised response-source tables.

For each endpoint, deployment-style fitting used the same 1,280 external plus all 85 ARROW labels, weights, preprocessing, architecture and seeds; only the 15 response columns differ. Tier-A labels were never used for fitting, teacher or feature selection, thresholding or model choice. The primary paired analysis and bootstrap convention match the confirmatory analysis. Complete eligibility decisions, source-exposure matches and molecule-level predictions are released as Supplementary Data 5. Because the compilation is external and molecular identities are disjoint, we call this an external molecule-disjoint cohort; we do not claim that all historical measurement provenance is independent.

Statistics

MAE in kcal mol^{-1} is primary. RMSE, median absolute error and the fraction of molecules improved are secondary. All comparisons are paired by molecule. Ninety-five per cent percentile intervals use 100,000 molecule bootstrap resamples with seed 20260828. For repeated partitions, each molecule's absolute error is first averaged across repeats before resampling. A predeclared comparison is positive when the complete paired interval for candidate-minus-control MAE is below zero, negative when above zero and neutral otherwise; an absolute difference of at least 0.010 kcal mol^{-1} is additionally termed material. No multiplicity correction or post-result feature selection is used.

Deployment artifact

After evaluation was frozen, the response teachers were fitted with standardized benchmark equivalents removed and the endpoint was refitted on the 1,280 public plus 85 ARROW labels. This all-data artifact is used only for deployment, never for paper accuracy. Inference parses the supplied SMILES with RDKit and converts it to canonical isomeric SMILES before computing descriptors and evaluating the six frozen response-surrogate artifacts; three endpoint regressors are averaged. This step is not a claim of invariance to tautomer, protonation or salt form. The runtime does not read an experimental value, family label, scaffold, fold identifier, trajectory or PIMD result. Software versions, hashes and a SMILES-only smoke test are reported in Supplementary Methods and Tables 7–8.

Data availability

All benchmark records, held-out predictions, split assignments, audit tables and derived source manifests required to reproduce the paper are provided in the SolvAI repository. Large third-party datasets are not redistributed where their size or licence makes that inappropriate; immutable URLs or DOIs, hashes, licences and exact preprocessing instructions are supplied in `repro/DATA_PROVENANCE.md`.

Code availability

Source code, the packaged inference artifacts, confirmatory analysis scripts, figure generation, manuscript source and a locked environment are available at https://github.com/Scientific-Computing-Lab/SolvAI. A frozen-artifact path reproduces predictions, metrics, figures and PDFs without rerunning physical calculations.

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